

BEHAVIORAL CONTAGION IN SIBSHIPS

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(Received 16 January 1991; revised 9 August 1991)

Summary—A behavior is “contagious” if one person is more likely to exhibit it when a relevant other person has already done so. In this sense, behavioral contagion is commonly thought to contribute to many social problems, such as drug abuse and teenage promiscuity. In this paper we focus on behavioral contagion in sibships. Borrowing a model from the theory of contagious diseases, we show that contagion will cause prevalence to increase with sibship size. This model also allows us to estimate the magnitude of the contagious factor relative to non-contagious factors. Finally, we develop two statistical tests for the presence of contagion. Results are presented for participation in a skill-development program and four child-psychiatric conditions: neurosis, hyperactivity, somatization, and conduct disorder. Evidence is presented that program participation is probably contagious and conduct disorder possibly so. The other three child-psychiatric conditions are shown not to be contagious. Implications for research and practice are discussed.

Introduction

A BEHAVIOR is “contagious” if one person is more likely to exhibit it when a relevant other person has already done so. In this sense, behavioral contagion is commonly thought to contribute to many social problems, such as drug abuse and teenage promiscuity. Determining the extent to which a behavior is contagious is problematic, however, because one cannot observe the reasons why a person exhibits a behavior. But contagion does have one directly observable consequence: the proportion of a group which exhibits a behavior (prevalence) tends to increase with group size.

Table 1 presents two non-behavioral examples of this tendency. For two contagious diseases, respiratory illnesses and gastroenteritis, the number of illnesses per person per year increases regularly with family size. Note that this increase is not just a matter of large families having more illnesses than small ones; they do, of course. Table 1 documents a much stronger proposition, namely, that the number of illnesses *per person* per year increases with family size. Intuitively, the larger the family, the more likely it is that at least one member will “bring home” the infectious agent. Hence, members of large families are more likely to be exposed to the disease than members of small ones.

This association between prevalence and family size allows us to detect contagion in a cross-sectional survey. In contrast, earlier researchers attempted to detect contagion by observing temporal patterns in behavior; that is, if person A exhibited some behavior and person B did so soon afterwards, person B would be said to have been influenced by A

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Table 1
Incidence of common respiratory diseases and infectious gastroenteritis by family size

Family size ^a	Illnesses/person/year	
	Respiratory diseases	Gastroenteritis
3	4.5	0.97
4	5.2	1.18
5	5.5	1.53
6	6.3	1.89
7	6.7	1.89
8	—	2.11

Note. From *Family Medicine Principles and Applications* by J. Medalie, 1978, Baltimore: Williams and Wilkins Company. Copyright 1978 by J. H. Medalie. Adapted by permission.

^aThe term "family" in this table includes the parents as well as their children.

(Polansky, Lippitt, & Redl, 1950; Grosser, Polansky, & Lippitt, 1951; Lippitt, Polansky, Redl, & Rose, 1952). This approach, however, requires continuous observation. Moreover, as Polansky et al. (1950) showed, different observers may disagree as to whether a person's behavior is the result of contagion.

Admittedly, detecting contagion via the relationship of prevalence and family size is indirect. But there are not many alternative mechanisms that can mimic a tendency for prevalence to increase with family size. Moreover, as will be seen, these alternative mechanisms can often be ruled out by appropriately designed statistical tests.

In this paper we focus on behavioral contagion in sibships. Borrowing a mathematical model from the theory of contagious diseases, we show that contagion will cause prevalence to increase with sibship size. This model also allows us to estimate the magnitude of the contagious factor relative to non-contagious factors. Finally, we develop two statistical tests for the presence of contagion.

Results are presented for participation in a skill development program and four child-psychiatric conditions: neurosis, hyperactivity, somatization, and conduct disorder. The skill-development program was carried out in Ottawa, Ontario, from 1980 to 1983, and the data for the child-psychiatric conditions were provided by a recent province-wide survey in Ontario. Evidence is presented that program participation is probably contagious and conduct disorder possibly so. The other three child-psychiatric conditions are shown not to be contagious. Implications for research and practice are discussed.

Methods

Theoretical Model

Children at risk for a given behavior are usually concentrated in certain age ranges and frequently in one or the other sex. We will refer to siblings within the appropriate age and sex categories as "susceptible." The number of susceptible siblings in a sibship will be called its "effective size." "Prevalence" is the fraction of susceptible siblings who exhibit a behavior. Our key proposition is that prevalence should increase with effective sibship size.

To establish this proposition, we will borrow a mathematical model developed by Longini and Koopman (1982) to explain the spread of contagious diseases. In this model, individuals may become infected from either the community or the household. The risk of infection from the community is the same for everyone. The risk of infection from the household, however, depends on the number of infected individuals in the household and therefore varies from family to family.

Longini and Koopman assume that initially no one had the disease. For any individual, the probability of avoiding infection from the community over the entire observation period is assumed to be constant B . The probability of avoiding infection from an infected member of one's household is assumed to be a constant Q . Infectious contacts are assumed independent, so that the probability of avoiding contact with j infected household members is Q^j .

Under these assumptions, Longini and Koopman derive a formula for the probability, denoted m_{jk} , that there will be j finally infected individuals in a household with k initial susceptibles. For households with only one susceptible the formula is trivial since infection can only come from the community. The individual avoids infection with probability $m_{01} = B$ and is infected with probability $m_{11} = 1 - B$.

For households with two susceptibles the formulas are still relatively simple. In order for there to be no cases, both susceptibles must avoid infection from the community; hence, $m_{02} = B^2$. In order for there to be exactly one case, one susceptible must be infected from the community (probability = $1 - B$) and the other must avoid infection from both the community and the infected sibling (probability = BQ). Since there are two choices for which susceptible is infected, we have $m_{12} = 2(1 - B)BQ$. Finally, since there must either be zero, one, or two cases, the probability that there are two cases can be computed as $m_{22} = 1 - m_{02} - m_{12}$.

For households with $k > 2$ susceptibles the formulas become more complex. However, it is possible to express them in the following recursive form:

$$m_{0k} = B^k \quad (1)$$

$$m_{jk} = \binom{k}{j} B^{k-j} Q^{j(k-j)} m_{jj}, \quad \text{for } 1 \leq j \leq k-1, \quad (2)$$

$$m_{kk} = 1 - \sum_{j=0}^{k-1} m_{jk}. \quad (3)$$

To use these formulas, one iteratively computes the values of m_{jk} starting with $k = 1$.¹

Prevalence is defined as the number of infected individuals in a household divided by

¹A brief explanation of equation (2) is as follows. Each unaffected susceptible has to avoid infection both from the j infected individuals (probability = Q^j) and from the community (probability = B); the probability that $k-j$ susceptibles avoid infection is therefore $(BQ^j)^{k-j} = B^{k-j} Q^{j(k-j)}$. The remaining j susceptibles which *do* catch the disease either catch it from the community or from each other; in effect, they act like a household of size j in which all j are infected, and the probability of this happening is m_{jj} . Finally, we must multiply by the number of ways of choosing the j infected individuals from the k initial susceptibles.

the number of initial susceptibles. For the Longini–Koopman model, the expected prevalence for a household with k initial susceptibles, P_k , is given by

$$P_k = \frac{1}{k} \sum_{j=0}^k j m_{jk}. \quad (4)$$

To illustrate this function Figure 1 plots expected prevalence versus the number of susceptibles for $B=0.8$ and for several different values of Q . The case $Q=1$ corresponds to there being no contagion (each susceptible avoids infection from household members with a probability of 1). In this case, prevalence is constant. When there is contagion, however, prevalence increases with effective sibship size; moreover the stronger the contagious effect (the smaller Q), the more prevalence increases.

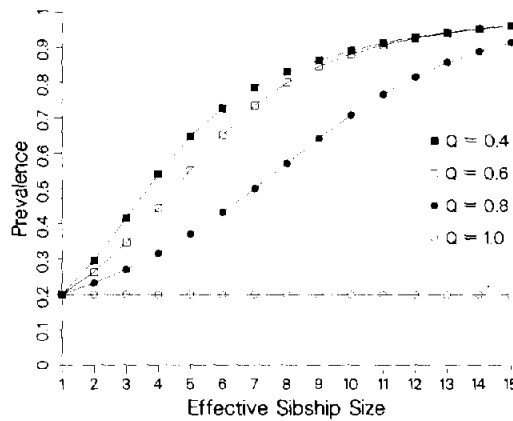


Figure 1. Expected prevalence as a function of effective sibship size for the Longini–Koopman model with $B=0.8$ and various values of Q .

Although contagion ($Q < 1$) causes prevalence to increase with effective sibship size in Figure 1, the figure does not constitute a proof. A formal proof is given in the Appendix. This proof is quite general and does not rely on the specific form of the Longini–Koopman model.

Estimating Parameters in the Longini–Koopman Model

Longini and Koopman (1982) detail how the parameters B and Q can be estimated by maximum likelihood. The data for the estimation consist of the effective size and number of affected children for all sibships under study. For any values of j and k ($j \leq k$), one counts the *observed* number of sibships with j out of k children affected. The *predicted* number of sibships with j out of k children affected is just $\hat{m}_{jk}$ times the total number of sibships of size k (where $\hat{m}_{jk}$ is the value of m_{jk} for the estimated values of B and Q). Intuitively, the estimation procedure chooses $\hat{B}$ and $\hat{Q}$ to maximize the fit between these observed and predicted values. The procedure also provides asymptotic standard errors

for the two parameters. An implementation of the estimation procedure for the IBM PC (or compatible machines) is available from the senior author (D. R. Jones).

Given the estimated parameters and their standard errors, one could develop a statistical test for contagion, that is, a test for whether $\hat{Q} < 1$. There are, however, some technical difficulties with this approach.² Moreover, in applying this test, one would not only be testing for contagion but also for the exact form of the Logini-Koopman model. As shown in the Appendix, however, contagion in almost any form will cause prevalence to increase with sibship size. We prefer, therefore, to develop tests for significance based on the relationship between prevalence and sibship size.

Testing for Presence of Behavioral Contagion

The preceding section showed that if contagion exists, one expects observed prevalence to increase with effective sibship size. The next step is a formal test that any such observed increase is statistically significant. The statistical theory that applies to this case involves testing a null hypothesis that several proportions are equal against the alternative that they increase monotonically. This theory has been extensively studied by Barlow, Bartholomew, Bremner, and Brunk (1972) and Schaafsma (1966). Following these authors, we construct a weighted sum, T , of the differences $P_{i+1} - P_i$ for $i = 1, 2, \dots, I-1$ (where I is the largest effective sibship size). If there is no behavioral contagion, all of the prevalences are expected to be the same and the value of T should be close to 0. If there is contagion, the prevalences are expected to increase ($P_{i+1} > P_i$) and the value of T should be positive. Intuitively, then, one should be able to reject the null hypothesis of no contagion if T is a sufficiently large positive number.

More formally, one computes the statistic $T' = T/s_T$ where s_T is an estimate of the standard error of T . Under the null hypothesis of no contagion, T' will be approximately normally distributed with a mean of 0 and unit variance. We can therefore reject the null hypothesis (and accept the alternative of behavioral contagion) using a standard one-tailed test based on T' . That is, one can reject the null hypothesis at the α -level if $T' > \xi_\alpha$ where ξ_α is the $1-\alpha$ quantile of the standard normal distribution. Details of the T' test can be found in the Appendix.

Excluding Other Explanations for Increasing Prevalence

Many mechanisms other than behavioral contagion can explain a tendency for prevalence to increase with effective sibship size. Most of these alternative mechanisms posit that large families tend to have some property (lower social class, less parental supervision, etc.) and that this property is associated with the behavior in question (e.g. conduct disorder). These alternative explanations differ from contagion in one key respect. Whereas behavioral contagion implies that prevalence will be related to the number of susceptible children (effective sibship size), the alternative explanations imply that prevalence will be related to the *total* number of children, both susceptible and non-susceptible.

²The problem is that, under the null hypothesis, Q lies on the boundary of the parameter space. Under such conditions, the distribution of the standard likelihood ratio statistic is no longer chi-squared but rather a mixture of chi-squared distributions; see Self and Liang (1987) for details.

Consider, for example, one family with three teenage children and another with two teenage children and an infant. All of the teenagers, but not the infant, are at risk from smoking. Under a hypothesis of behavioral contagion, we would expect the prevalence of smoking (i.e., the proportion of *teenagers* who smoke) to be greater in the family with three teenagers than in the family with just two. Under most explanations based on total sibship size, however, expected prevalence would be the same since the two families have the same total size. It follows that most explanations based on total sibship size can be excluded if prevalence can be shown to increase with the number of susceptibles even when total sibship size is held constant.

The foregoing considerations lead naturally to a formal test which can distinguish behavioral contagion from alternative explanations based on total sibship size. The basic idea is simple: divide the data into groups with the same total sibship size, and perform separate T' tests on each group. Since total size is constant within groups, any tendency for prevalence to increase with *effective* size (i.e., positive T') can be attributed to contagion as opposed to a total size effect.

These separate T' tests can be combined into a single, overall test for contagion that controls for total size. One first forms a sum, U , of the T statistics for each total size. If there is no contagion, then each T statistic has a mean of 0 and so U should be near 0. If there is contagion, however, then each T has a positive mean and U should be positive. Intuitively, then, one should reject the null hypothesis of no contagion if U is a sufficiently large positive number.

More formally, one computes the statistic $U' = U/s_U$ where s_U is an estimate of the standard error of U . Under the null hypothesis of no contagion, U' will be approximately standard normal. We can therefore test the null hypothesis using a standard one-tailed test. Details of the U' test can be found in the Appendix. An implementation of the T' and U' tests for the IBM PC is available from the senior author (D. R. Jones).

Although useful, the U' test does not exclude all alternative explanations for increasing prevalence. For example, suppose young children are the susceptible individuals for some behavior. Suppose, further, that families from disadvantaged ethnic groups have a higher proportion of young children than established groups. It follows that, for fixed total sibship size, the number of susceptible (young) sibs would be positively correlated with membership in these disadvantaged groups. If membership in disadvantaged ethnic groups is correlated with the behavior, then prevalence would be positively correlated with effective sibship size—and for reasons completely unconnected with contagion. (We thank one of the reviewers for this example.)

In general, the U' test controls for factors that are correlated with total sibship size but *uncorrelated* with effective size (at least when total size is controlled for). The previous example did not fit this description because the factor "membership in disadvantaged ethnic groups" was correlated mainly with effective sibship size (number of young children) and not total size. However, factors such as ethnic membership can be taken into account by modifying the U' test. Instead of dividing the data into groups based on total size, one divides it into groups based on ethnic origin. Everything else then remains the same: we compute separate T statistics for each group and combine them into an overall U' statistic.

Study 1: Program Participation

Introduction

"Participate and learn skills" (PALS) was a skill-development program available to all children 5–15 years old living in a publicly supported housing complex in Ottawa, Ontario (Offord & Jones, 1983; Jones & Offord, 1986). During 1980 PALS sponsored 40 programs in 26 different skill areas. All of the children in the complex were informed about PALS programs. Lists were kept of who participated and who did not, and special efforts were made to reach non-participating children. The only form required was one giving parental permission, and no fee was charged.

Subjects and Procedures

In 1980 there were 181 families which contained children eligible for PALS and which were resident in the housing complex throughout the year. These 181 families contained 322 eligible children (i.e., children 5–15 years old). Of these children, 228 participated in PALS, where "participation in PALS" meant attending at least one session of any program. This rate of participation (71%) was very good for a poor community, especially in a program's first year.

Families who lived in the housing complex throughout 1980 were identified from records kept by the Ottawa Housing Authority (OHA). These records provided information as to children in the family, their birthdates, and sexes.

Results

Table 2 presents the 181 sibships in the all-year 1980 sample classified by effective sibship size. Number of eligible children, number of participating children, and prevalence are also presented. The value of T' equals 4.18, which is significant well beyond the 0.001 level. We may conclude that the prevalence of participation does increase with effective sibship size.

Table 2
Sibships in the Ottawa study classified by effective sibship size

Effective sibship size	Number of sibships	Number of eligible children	Number of participating children	Prevalence (%)
1	101	101	64	63.4
2	42	84	50	59.5
3	22	66	48	72.7
4	9	36	33	91.7
5	7	35	33	94.3
Total	181	322	228	

Several of the families in the PALS study had children who were either less than 5 years old or more than 15 but living at home. Since these children could not participate in PALS,

they were not susceptible. The presence of such non-susceptibles means that effective size can vary independently of total size. Table 3 presents prevalence cross-classified by effective and total sibship size. The value of U' for these data is 1.07 which is only significant at the 0.14 level. Thus, when we control for total sibship size, the data are not strong enough to establish a tendency for prevalence to increase with effective sibship size.

Table 3
Sibships in the Ottawa study cross-classified by effective and total sibship size

Effective sibship size	Total size 2		Total size 3		Total size 4		Total size 5	
	<i>N</i>	<i>P</i> (%)	<i>N</i>	<i>P</i> (%)	<i>N</i>	<i>P</i> (%)	<i>N</i>	<i>P</i> (%)
1	10	0.40	3	0.67	—	—	—	—
2	68	0.54	14	0.79	—	—	2	1.00
3	—	—	60	0.73	6	0.67	—	—
4	—	—	—	—	32	0.91	—	—
5	—	—	—	—	—	—	20	0.95

Note. *N* is the number of susceptible children on which the corresponding prevalence *P* is based. Data for sibships of total sizes 1, 6, 7, and 8 are not shown because the T' test could not be completed (there was only one effective size).

This lack of significance in the U' test is partially due to the small number of ineligible (non-susceptible) children in the PALS data. Only 27 of the 181 families had ineligible sibs. As a result, prevalences in Table 3 for effective sizes less than the total size are based on small samples. These small sample sizes make it difficult to find *significant* differences between prevalences. The results therefore serve to underscore the value of having substantial numbers of sibships with and without non-susceptible sibs.

Finally, the Longini-Koopman model was fit to the Ottawa data obtaining estimates of $\hat{B} = 0.43$ and $\hat{Q} = 0.58$ (with asymptotic standard errors of 0.036 and 0.061, respectively). The fact that $\hat{Q}$ is less than 1 indicates contagion; given a participating sib in the family, other eligible sibs have a 42% chance ($1 - \hat{Q}$) of being influenced by him. In addition, each sib has a 57% chance ($1 - \hat{B}$) of participating in PALS due to general community influences. Table 4 compares the actual versus predicted number of sibships cross-classified by the number of eligible and participating siblings. As one can see, the fit is quite good.

Study 2: Conduct Disorder

Introduction

A sizable literature supports the proposition that conduct-disordered boys are more likely to come from large families than comparable boys who are not conduct disordered (Glueck & Glueck, 1950; Hirschi, 1969; Nye, 1958; Robins, West, & Herjanic, 1975; Ryan, 1974; Shanok & Lewis, 1977; West & Farrington, 1973). The difference appears to hold for both officially recognized and self-reported behaviors and, furthermore, to remain after statistical account is taken of income, socioeconomic status, and parental criminality (Fischer, 1984). It has been recognized, moreover, that such a difference can be explained by behavioral contagion. "It looks," write Robins et al. (1975), "as though a principal mechanism through

Table 4
Actual versus predicted number of sibships in the Ottawa study classified by number of eligible and participating siblings

Number of eligible children	Number of participating children	Number of sibships	
		Actual	Predicted
1	0	37	43.3
1	1	64	57.7
2	0	12	7.7
2	1	10	11.9
2	2	20	22.4
3	0	3	1.7
3	1	2	2.3
3	2	5	5.0
3	3	12	12.9
4	0	0	0.3
4	1	0	0.3
4	2	0	0.6
4	3	3	1.7
4	4	6	6.1
5	0	0	0.1
5	1	0	0.1
5	2	0	0.1
5	3	0	0.3
5	4	2	1.1
5	5	5	5.3

which large families encourage delinquency may be through increasing the risk of exposure to a delinquent sibling.”

Subjects and Procedures

The Ontario Child Health Study (OCHS) was a province-wide survey, conducted in 1983, of the psychiatric health of children aged 4–16 (Boyle et al., 1987; Offord et al., 1987). The target population was all children living in Ontario who were born between 1966 and 1978, inclusive. Only three groups of children, representing 3.3% of the population aged 4–16, were excluded: those living on Indian reserves, in institutions, or in dwellings constructed since June 1, 1981. Statistics Canada selected a stratified random sample of households from the 1981 census file and 91% of these households selected agreed to participate in the study, yielding a total of 3,294 children. Survey instruments included household record cards (to screen households for eligible children), interviewer-administered questionnaires, a parent self-administered questionnaire, and self-report checklists. With the exception of the household record cards, these instruments focused on four childhood disorders: neurosis, hyperactivity, somatization, and conduct disorder.

Results

Table 5 presents prevalence by effective sibship size for the four conditions. Susceptible children were defined so as to include all children 4–16 for neurosis, all boys 4–16 for

Table 5

OCHS prevalences by effective sibship size for neurosis, hyperactivity, somatization, and conduct disorder, where susceptible children are defined to include children 4-16 (neurosis), boys 4-16 (hyperactivity), girls 12-16 (somatization), and boys 8-16 (conduct disorder)

Effective sibship size	Neurosis		Hyperactivity		Somatization		Conduct disorder	
	N	P(%)	N	P(%)	N	P(%)	N	P(%)
1	679	9.4	331	9.1	471	8.5	638	7.5
2	1200	10.5	613	0.8	134	2.2	314	9.2
3	584	9.4	312	5.4	21	0.0	72	9.7
4	160	10.6	77	3.9	—	—	4	25.0
5	48	4.2	24	8.3	—	—	—	—
6	10	0.0	5	0.0	—	—	—	—
7	7	14.3	3	0.0	—	—	—	—

Note. *N* is the number of susceptible children on which the corresponding prevalence is based. The numbers for neurosis and hyperactivity are not always multiples of sibship size because of missing cases. The data for somatization and conduct disorder are, however, complete. *P* (prevalence) is the fraction of susceptible children (for whom we have data) who are affected.

hyperactivity, girls 12-16 for somatization, and boys 8-16 for conduct disorder. The age ranges for susceptibility were limited in the first instance by the OCHS, which only sampled children 4-16 years old. Hyperactivity was limited to boys because in the OCHS, as in other surveys, it affects boys primarily. Somatization was limited to adolescent girls because it is more than four times as prevalent among them as in any other group (OCHS data). Conduct disorder is known to affect boys primarily, beginning with the early school years. Of these behaviors, only the prevalence of conduct disorder shows a tendency to increase. The value of T' for conduct disorder is 1.53, which yields a significance level of 0.06.

Table 6 shows prevalence for conduct disorder cross-classified by effective and total sibship size. For total sibships sizes 2 and 5, prevalence shows some tendency to increase with effective size, but the results for total sibship sizes 3 and 4 are unclear. The value of U' for these data is 0.86 which is only significant at the 0.19 level. Thus, when we control for total sibship size, the data are not strong enough to establish a convincing case for contagion.

Table 6

Prevalence of conduct disorder among OCHS children classified by effective and total sibship size

Effective sibship size	Total size 2		Total size 3		Total size 4		Total size 5	
	N	P(%)	N	P(%)	N	P(%)	N	P(%)
1	302	0.07	142	0.11	42	0.12	8	0.00
2	136	0.11	124	0.02	36	0.31	14	0.07
3	—	—	33	0.12	27	0.04	6	0.33
4	—	—	—	—	—	—	4	0.25
5	—	—	—	—	—	—	—	—

Note. *N* is the number of susceptible children on which the corresponding prevalence *P* is based. Data for sibships of total size 1 are not shown because the T' test cannot be computed (there is only one effective size). Data for sibships of total sizes 6 and 7 are not shown because there are no affected children; as a result $T - s_T = 0$ and these sibships do not contribute to the U' statistic.

The estimated parameters in the Longini-Koopman model are $\hat{B} = 0.922$ and $\hat{Q} = 0.886$ (with asymptotic standard errors of 0.008 and 0.059, respectively). If a conduct-disordered sibling is present in the household, each susceptible child has an 11.4% chance ($1 - \hat{Q}$) of being influenced by him (and becoming conduct disordered). This compares to a 7.8% chance ($1 - \hat{B}$) of becoming conduct disordered due to general community influences. Table 7 compares the actual versus predicted number of sibships classified by number of susceptible and affected children. Once again, the fit is quite good.

Table 7
Actual versus predicted number of OCHS sibships classified by number of children susceptible to and affected by conduct disorder

Number of susceptible children	Number of affected children	Number of sibships	
		Actual	Predicted
1	0	590	588.0
1	1	48	50.0
2	0	132	133.3
2	1	21	20.1
2	2	4	3.5
3	0	19	18.8
3	1	3	3.8
3	2	2	1.2
3	3	0	0.3
4	0	0	0.7
4	1	1	0.2
4	2	0	0.1
4	3	0	0.0
4	4	0	0.0

Of the four OCHS conditions only conduct disorder showed any evidence of contagion and, even in its case, the increasing trend of prevalence with sibship size reached only borderline significance. The magnitude of the contagious effect was small. In the absence of contagion one would expect 7.8% of the boys ($1 - \hat{B}$) to be conduct disordered. Since the observed percentage was 8.3%, it would appear that only 0.5% of the boys were susceptibles who became affected through contagion. In other words, conduct disorder can be attributed to behavioral contagion in only $(0.5/8.3 =)$ 6% of the boys whom it affects. It should be remembered that this result applies to this particular population of sibship sizes. If there had been more large sibships, the proportion of children affected via contagion would probably have been larger. It should also be remembered that most boys in the OCHS survey were only partway through the "window of susceptibility" (defined here as ages 8-16). Thus, some boys counted as unaffected in the survey may actually have become affected by the age of 16. These additional, uncounted infections would come partly from the community and partly from behavioral contagion. It follows that our results probably understate the true risk of both community infection ($1 - B$) and contagious infection ($1 - Q$).

Finally, it should be remembered that the OCHS results apply only to behavioral

contagion within sibships. It says nothing at all about the existence or extent of behavioral contagion in groups other than sibships. It is perfectly possible, for example, for conduct disorder to be weakly contagious within sibships and strongly contagious in teenage male friendship groups.

Implications for Research and Practice

“Familial clustering” is a tendency for a behavior (or any other characteristic) to be concentrated in certain families. More precisely, the proportion of families in which either all or none of the children exhibits the behavior is larger than would be expected if the behavior were randomly distributed. Such familial clustering is a frequent finding in psychiatric research. It can be caused by a variety of factors, including both behavioral contagion and genetic commonality.

To understand how behavioral contagion can lead to familial clustering, consider the extreme case when a behavior is completely contagious. That is, once a sibling brings home the behavior (from general community influences), all the rest of the children in the household exhibit the behavior. In this case there are only two possible outcomes: either none of the children exhibit the behavior or all of them do. The middle possibility (some but not all exhibiting it) is underrepresented and the two extremes overrepresented. In short, there is familial clustering.

Table 8 illustrates familial clustering with data from the PALS program. There were 42 sibships of effective size 2 and, hence, 84 susceptible children. Of these children 50, or 59.5%, participated in the skill-development program. Now if each child had a probability $P = 0.595$ of exhibiting the behavior, and if there were no contagion, the probability of both children exhibiting the behavior would be $P^2 = 0.354$. Hence, the expected number of sibships with both children participating would be 42 times 0.354, or 14.9 sibships. As shown in Table 8, however, there were 20 sibships in which both exposed siblings participated, which is significantly greater than the 14.9 one would expect if the behavior were randomly distributed.

Table 8
Expected and observed numbers of sibships in the Ottawa study classified by effective sibship size and whether all, none, or some but not all children participated

Effective sibship size	Exp/obs ^a	Sibs participating		
		All	None	Some
2	Exp	14.9	6.9	20.2
	Obs	20.0	12.0	10.0
3	Exp	8.5	0.4	13.1
	Obs	12.0	3.0	7.0
4	Exp	6.4	0.0	2.6
	Obs	6.0	0.0	3.0
5	Exp	5.2	0.0	1.8
	Obs	5.0	0.0	2.0

^aExpected/observed.

Table 8 shows similar calculations for effective sibship sizes 3, 4, and 5. In general, if there are N sibships of size k , and if the prevalence is P , one expects NP^k sibships with all k children participating, $N(1 - P)^k$ sibships with none of the children participating, and $N[1 - P^k - (1 - P)^k]$ sibships with some but not all of the children participating. For sibship sizes 2 and 3 the observed numbers of sibships with all or none participating significantly exceed the expected numbers; by the exact binomial test, p equals 0.001 and 0.008, respectively. For sibship sizes 4 and 5 prevalence is so high that the expected and observed values are virtually forced into agreement (no significant difference).

If familial clustering can be caused by either genetic commonality or contagion, how can we determine which is operative? One of the merits of the model presented in this paper is that it provides a criterion for distinguishing between these potential causes. That criterion, of course, is whether or not prevalence increases with effective sibship size. With few exceptions, conditions of genetic origin are equally prevalent regardless of effective sibship size.³ Hence, a tendency for prevalence to increase with the number of children is strong presumptive evidence against a genetic origin and for contagion.

A second implication of the behavioral contagion model concerns practice. It is a common observation that those children who most need psychiatric help are often the least willing to accept or "comply" with it. Compliance, however, may be contagious within sibships, just as participation was in the Ottawa study. If compliance is sufficiently contagious, a child who could not be induced to accept help if approached directly might do so if a brother or sister were involved first. In effect, one recruits a non-target sib who then recruits the target child. In short, there may be cases in which one can use the contagiousness of a behavior as a tool for encouraging or discouraging its spread. Of course, before adopting such an approach one needs to know whether the behavior is contagious, and it is here that the methods described in this paper are useful.

We have focused here on behavioral contagion within sibships. The methods, however, are more general and could be applied to groups other than sibships if the appropriate data were available. With some modification, the methods could also be extended to capture the transmission of a behavior from parent to child.⁴ With respect to contagion, the model could play a role similar to that played by twin and adopted-away designs in psychiatric genetics. It leaves a great deal unanswered, specifically, the mechanism by which one person's behavior affects that of another; but, like the twin and adopted-away designs, it enables one to know how important a particular kind of causation is in the etiology of a psychiatric condition.

³One exception is hemolytic disease of the newborn. This condition occurs when an Rh-negative woman and an Rh-positive man have children. These children may be Rh-positive and, if they are, the woman may develop anti-Rh antibodies (today, this reaction is usually prevented by administering Rh immune globulin). These antibodies appear too late to affect the first child, but subsequent children may be exposed and develop hemolytic disease. Thus, no children can be affected in sibships of size 1 (0% prevalence); only the second child can be affected in sibships of size 2 ($\leq 50\%$ prevalence); only the second and third children can be affected in sibships of size 3 ($\leq 67\%$ prevalence); and so forth. The net result is a slight tendency for prevalence to increase with sibship size.

⁴Such an extension might be relevant in studying teenage pregnancy, where it is known that an adolescent girl is more likely to bear a child if her mother had been a teenage mother (Furstenberg, Levine, & Brooks-Gunn, 1990). One explanation for this is that the daughter uses her mother as a role model—a sort of one-way contagion from parent to child. Note, however, that this one-way type of contagion does not cause prevalence to increase with sibship size, which is why we have not considered it in this paper.

Acknowledgements—The authors thank Dr David R. Offord for permission to use data from the Ontario Child Health Study and Dr Michael H. Boyle for invaluable help in processing those data. We also thank Professor W. Schaafsma of the University of Groningen (Netherlands) for guidance in constructing the T' and U' tests.

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Appendix

Proof that Prevalence Increases with Sibship Size

Consider any sibship of size $k + 1$. Choose any individual at random, and let the random variable Y be 1 if this individual is affected and 0 otherwise. Similarly, let the random variable X denote the number of affected sibs among the remaining k . The total number of affected sibs is then $X + Y$, and expected prevalence is

$$P_{k+1} = \frac{E(X+Y)}{k+1} \quad (5)$$

where $E(\cdot)$ denotes the expectation operator.

The probability that the randomly chosen individual is affected is, by definition, P_{k+1} ; hence, $E(Y) = P_{k+1}$. With respect to $E(X)$, note first that all siblings have the potential of "infecting" each other. Suppose, however, that the originally chosen individual cannot "infect" the remaining k sibs. These remaining sibs would then act like a sibship of size k and would have an expected prevalence of P_k . In our previous notation, we would have $E(X)/k = P_k$ or $E(X) = kP_k$. In reality, of course, the originally chosen individual will be affected a fraction P_{k+1} of the time and, when affected, may communicate the behavior to the remaining sibs. Since this contagious influence can only increase the probability that the remaining sibs are affected, it follows that $E(X) > kP_k$. Substituting these results into equation (5), we get

$$\begin{aligned} P_{k+1} &= \frac{E(X+Y)}{k+1} \\ &= \frac{E(X) + E(Y)}{k+1} \\ &> \frac{kP_k + P_{k+1}}{k+1}. \end{aligned} \quad (6)$$

Rearranging terms in equation (6) then gives

$$P_{k+1} > P_k, \quad (7)$$

which is the desired result. This proof, it should be noted, is quite general; in particular, we did not use the specific form of the Longini-Koopman model (equations 1-3).

The T' Test

Suppose we have data on sibships of effective sizes $k_1 < k_2 < \dots < k_I$. Let n_i denote the total number of susceptible children in sibships of effective size k_i , and let P_i denote the prevalence of the behavior among these n_i children. The T statistic is defined by

$$T = \sum_{i=1}^{I-1} w_i (P_{i+1} - P_i), \quad (8)$$

where each w_i is a positive weight. Collecting coefficients on P_i , we may rewrite equation (8) as

$$T = \sum_{i=1}^I c_i P_i, \quad (9)$$

where

$$c_i = w_{i+1} - w_i \quad (10)$$

and where, in equation (10), it is understood that $w_0 = w_I = 0$. Under the null hypothesis of no contagion, the variance of T may be estimated by

$$s_T^2 = \left[\sum_{i=1}^I \frac{c_i^2}{n_i} \right] \left[\frac{\sum_{i=1}^I n_i P_i (1 - P_i)}{N - I} \right], \quad (11)$$

where $N = \sum_{i=1}^I n_i$. The T' statistic is then $T' = T/s_T$.

As for the weights w_i , Schaafsma (1966) suggests

$$w_i = \left[\frac{1}{N} \left(\sum_{j=1}^i n_j \right) \left(N - \sum_{j=1}^i n_j \right) \right]^{0.5} \quad (12)$$

The idea behind these weights is as follows. We are testing the null hypothesis of equal prevalences against the alternative that these prevalences increase. Since prevalence can increase in many ways, the alternative is a composite hypothesis. It turns out that no single set of w_i weights can maximize power against all simple hypotheses in the alternative (i.e., against all patterns of increasing prevalence). Thus, some other criterion must be used to determine the weights. Schaafsma (1966) suggested the weights in equation (12) because they make the resulting test "most-stringent, somewhere-most-powerful." It is "somewhere-most-powerful" in that it maximizes power against at least one simple hypothesis in the alternative. It is "most-stringent" in that it minimizes the maximum "shortcoming" of the test. For any particular simple hypothesis in the alternative, the shortcoming is defined as the difference in power between the T' test and a test designed to maximize power against this simple alternative.

The T' test is equivalent to test (5.19) of Schaafsma (1966). Readers who consult Schaafsma should be warned that our notation differs significantly from that used in his book.

The U' Test

For the U' test, we first divide the data into groups based on total sibship size. Although total size is fixed within groups, sibships can differ with respect to effective size due to the presence of non-susceptible sibs. As long as a group contains at least two effective sizes, we can compute a within-group T' test using the method of the previous section.

Let M denote the number of total sibship sizes for which we can compute a within-group T' test, and let the m th test ($m = 1, 2, \dots, M$) be given by $T'_m = T_m/s_{Tm}$. We then sum these separate T_m statistics to form an overall U statistic:

$$U = \sum_{m=1}^M T_m. \quad (13)$$

The variance of U is estimated by

$$s_U^2 = \sum_{m=1}^M s_{Tm}^2. \quad (14)$$

Finally, the U' statistic is U/s_U . From equation (13) it may appear that the T_m values are being summed without any appropriate weighting for sample size. Actually, the w_i values defined in equation (12) implicitly weight each T_m in an appropriate manner. For details, see Schaafsma (1966).